



CAPABILITY
PORTFOLIO
2026

REMOTE SENSING · EARTH OBSERVATION · APPLIED AI

Seeing Earth, Smarter

AI-powered Earth Observation for governments, agribusiness, climate finance
and geospatial partners from national mapping agencies to insurers.

RASID S.A.R.L. · BEIRUT, LEBANON

APP.RASID.AI · RASID.AI

Inside this Portfolio

Who we are, what we deliver, and a decade of funded work, six engagements in detail, followed by the full project register.

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A Boutique Geospatial & AI Consultancy

RASID S.A.R.L. turns satellite imagery into decisions for national mapping agencies, ministries, agribusiness, insurers and climate-finance institutions. We combine deep remote-sensing know-how with modern AI to deliver monitoring products that would otherwise take field teams years to produce.

01

~10 years

Team experience in remote sensing, deep learning and time-series satellite imagery, backed by an in-house Python processing library.

02

Multi-country

Legal entities in Lebanon, France and Egypt, serving clients across the Middle East, Africa and Europe.

03

Two platforms

A production SaaS (app.rasid.ai) and GoPilot, our AI geospatial agent, alongside bespoke consulting engagements.

\$700B

EO MARKET BY 2030

27.5%

AI-IN-EO CAGR

10+

FUNDED PROGRAMMES

04

~60 + journal papers

Strong in-house R&D team with 60+ journal & conference publications.

05

Winner of the AWS Geospatial Gen AI Challenge (August 2026)

Recognition for GoPilot, our AI geospatial agent in the AWS market.

Capabilities & Services

RASID delivers focused monitoring products and one-off analyses across imagery quality, urban and environmental change, climate and GHG, agriculture and document AI all on the same in-house pipeline.

01

Satellite Image Quality & Orthorectification

Anomaly detection (cloud, haze, terrain distortion, road shifts) plus geometric & radiometric correction.

02

Urban & Environmental Monitoring

Settlement growth, building change, road condition and urban heat-island analysis from multi-source imagery.

03

RASID Climate Action

Mitigation (landscape, crop and soil monitoring) and adaptation (heat islands, flood/wildfire risk, resilience dashboards).

04

Agriculture Analytics

CropWatch, BananaSight (TR4), national wheat mapping, grain-bin classification and yield forecasting.

05

AI Platforms — SaaS & GoPilot

Single-click, no-code AI analysis (app.rasid.ai) and GoPilot, a conversational AI geospatial agent.

06

RASID AI Shop

End-to-end AI consulting: MLOps (AWS/GCP/Azure), agentic AI, OCR, speech-to-text and text-to-speech.

Target sectors

Three groups of clients drive most of our engagements, each with distinct data, compliance and delivery needs.

01

URBAN · DEFENCE · INFRASTRUCTURE

Scene & environment monitoring

Urban and construction-site monitoring, road-network condition assessment (cracks and potholes), mining-site mapping, and geospatial intelligence & reconnaissance (GEOINT, S&R) for the defence sector.

02

AGRI-ANALYTICS · EUDR · VALUE: \$785.3M BY 2028

AI-based satellite imaging for agriculture

Crop-damage and pesticide-usage monitoring, agribusiness analytics, EU Deforestation Regulation (EUDR) compliance, and soil-quality & smart-fertilizer management.

03

MINISTRIES · OIL & GAS · CBAM · VALUE: \$3.28B (2022)

MRV for GHG emissions, weather & climate change

Monitoring, reporting and verification for ministries and local governments, the oil & gas and waste-management sectors, and the EU's Carbon Border Adjustment Mechanism (CBAM).

EBRD CLIMATE CHANGE INNOVATION PROGRAMME · 2025

MethaneMapper: Landfill Methane Plume Detection

ADD VISUALES :)

OVERVIEW

A satellite-based system to detect and quantify methane plumes from landfill sites using multi- and hyper-spectral imagery.

SOLUTION (METHODOLOGY)

Multi-modality AI techniques process satellite imagery to detect and quantify methane plumes, fine-tuned against ground-truth data.

PROBLEM

Landfill methane emissions are a major, under-monitored contributor to greenhouse gas output, with no consistent way to detect plumes at scale.

RESULTS

Delivered a working plume-detection pipeline under an EBRD Climate Change Innovation Programme grant of €27,000.

EBRD LEBANON INNOVATION PROGRAMME · 2025

BananaSight: Fusarium Wilt (TR4) Detection



OVERVIEW

Real-time monitoring for banana plantations to detect early signs of stress and disease using Sentinel-2 multispectral imagery.

PROBLEM

Fusarium Wilt (TR4) can devastate banana crops before visible symptoms appear, and farmers lack an early-warning tool.

SOLUTION (METHODOLOGY)

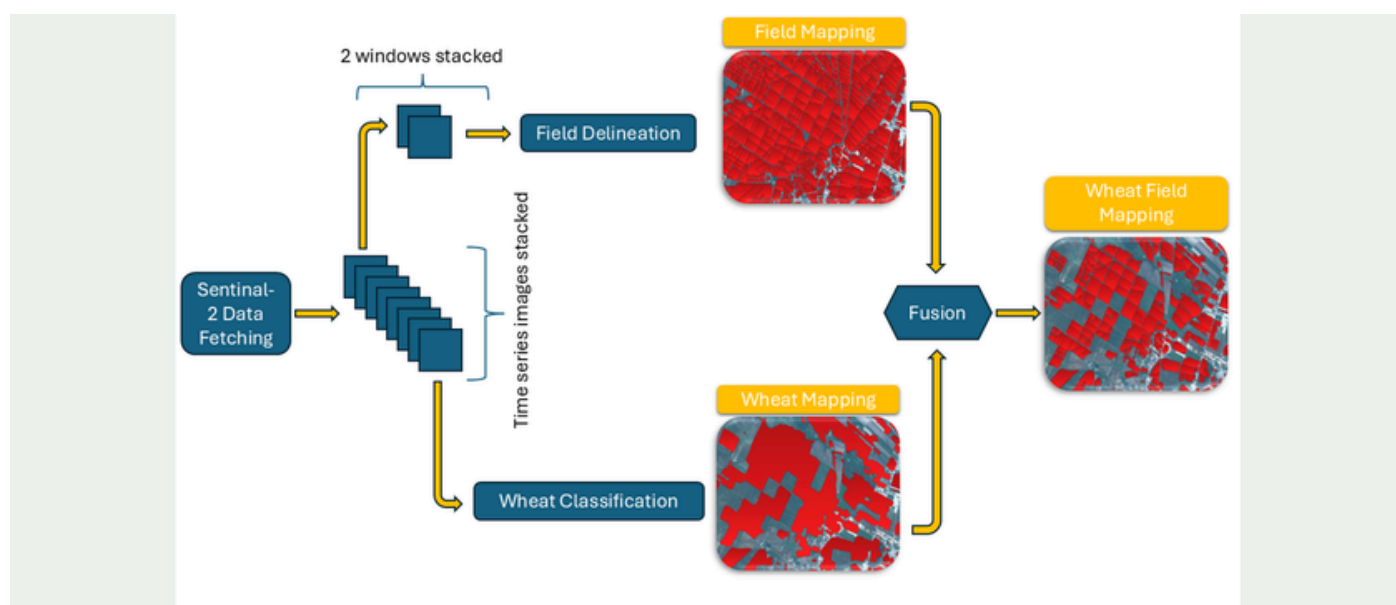
Sentinel-2 imagery is analysed to detect and alert farmers of potential stressors and disease signatures ahead of visible outbreak.

RESULTS

First commercially-ready multispectral solution of its kind, delivered under an EBRD Lebanon Innovation Programme grant of €30,000.

ESA NETWORK OF RESOURCES · 2024–2025

Winter Wheat Detection from Sentinel-2 Imagery



OVERVIEW

National-scale wheat-field delineation and tracking using a transformer-based foundation model on Sentinel-2 time-series imagery.

PROBLEM

Governments and traders lack up-to-date, parcel-level wheat maps for food-security planning and yield forecasting.

SOLUTION (METHODOLOGY)

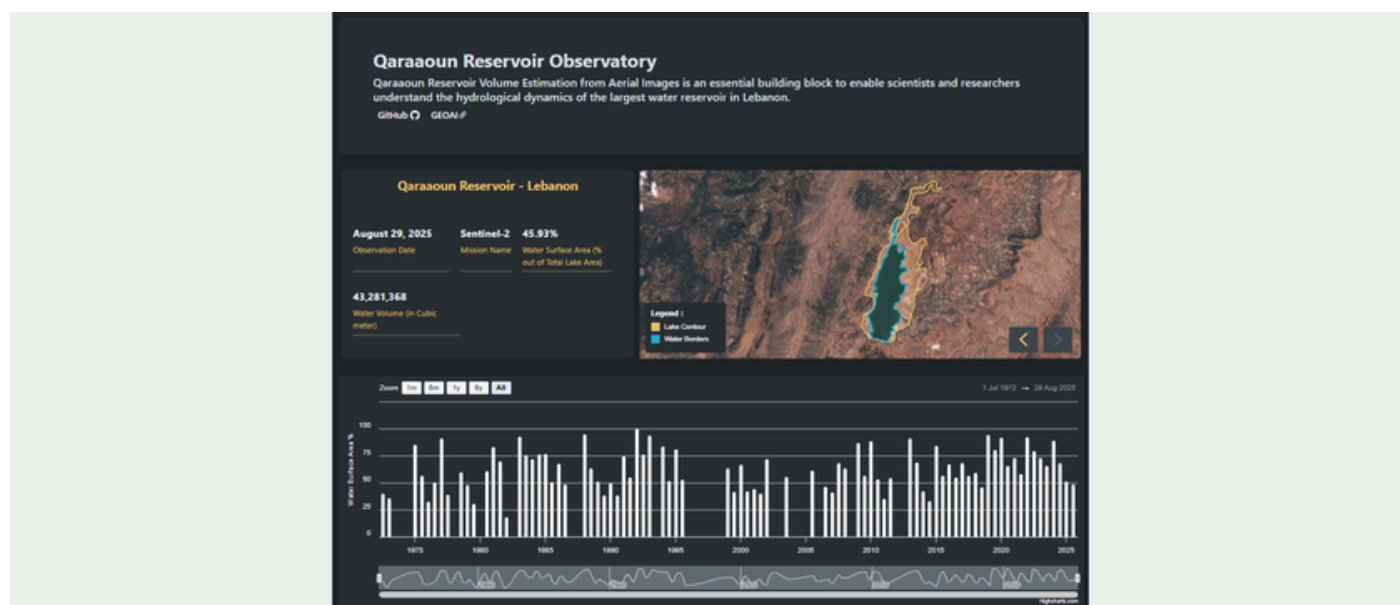
A transformer-based foundation model processes time-series optical and SAR imagery to delineate and classify wheat fields at national scale.

RESULTS

Produced a decoded winter-wheat map for Lebanon under an ESA Network of Resources grant of €5,000.

SEALACOM RESEARCHERS · 2023–2024

Qaraaoun Reservoir Water Monitoring



OVERVIEW

Earth-observation-based monitoring of reservoir extent and water levels for Lebanon's Qaraaoun Reservoir.

PROBLEM

Manual field surveys of reservoir water levels are slow and infrequent, limiting timely water-resource decisions.

SOLUTION (METHODOLOGY)

Satellite imagery is processed to track reservoir surface extent and water-level change over time.

RESULTS

Delivered a water-monitoring dataset for the Qaraaoun Reservoir under a SEALACOM Researchers grant of €4,500.

EBRD LEBANON INNOVATION PROGRAMME · 2023–2024

Buildings' Rooftop Segmentation

ADD VISUALES :)

OVERVIEW

Deep-learning segmentation of building rooftops from very-high-resolution satellite imagery.

PROBLEM

Rooftop inventories for energy and urban-planning use cases are typically outdated or incomplete.

SOLUTION (METHODOLOGY)

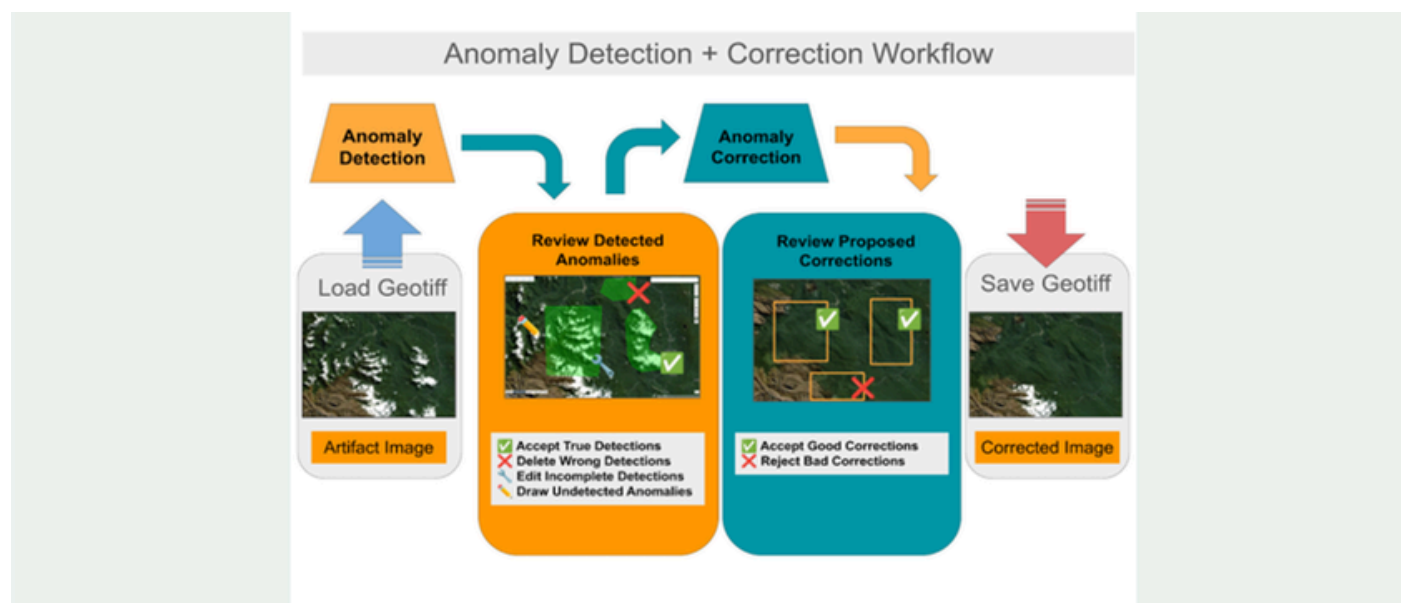
A deep-learning segmentation model identifies individual building rooftops from tasked satellite imagery.

RESULTS

Delivered a rooftop layer used to update Lebanon's solar rooftop potential map, under an EBRD grant of €5,000.

PRIVATE SATELLITE-IMAGERY PROVIDER · 2026

C-ARD: Corrected Analysis Ready Data



OVERVIEW

An AI pipeline that automatically corrects geometric and radiometric anomalies in high-resolution satellite imagery, turning raw data into Analysis-Ready Data (ARD) and refining existing ARD to remove residual errors.

SOLUTION (METHODOLOGY)

Deep-learning co-registration & orthorectification, radiometric enhancement, and automatic cloud/haze/shadow detection & removal, with a human-in-the-loop GUI — deployable in cloud, on-prem, or on-orbit (EC-ARD).

PROBLEM

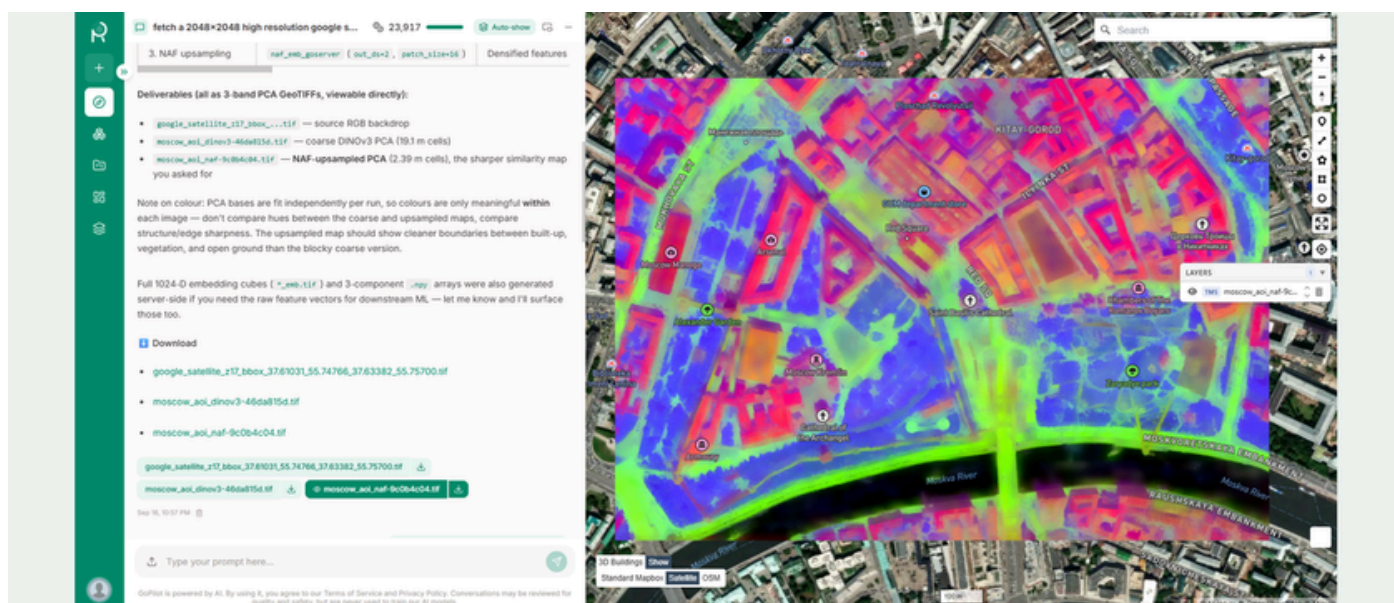
Providers battle geometric distortion, inconsistent radiometry and atmospheric interference (cloud, haze, shadow) — driving high rejection rates, while manual correction is slow, costly and unscalable.

RESULTS

A client-ready ARD pipeline processing ~200 GB in ~2 hours — turning more captures into sellable imagery, cutting QA cost and accelerating delivery for a private imagery provider.

AWS BEDROCK · FLAGSHIP PLATFORM

GoPilot RASID's AI Geospatial Agent



OVERVIEW

A conversational AI agent that puts RASID's remote-sensing stack in front of analysts through plain-language requests.

SOLUTION (METHODOLOGY)

Built on AWS Bedrock and the Strands framework with Claude as the LLM backbone, using SAM3/DINOv3 vision pipelines, delivered via a QGIS plugin and ArcGIS Pro add-in.

PROBLEM

Geospatial analysis workflows require specialist GIS skills and manual scripting for every new task.

RESULTS

Winner of the AWS Geospatial Gen AI Challenge (Aug 2026); piloted with CIGN (Côte d'Ivoire) and OSGOF (Nigeria).

Funded Projects, Grants & ToRs

A decade of small and mid-size funded engagements with agencies including ESA, the EBRD and the AOAD each one a validated, delivered use case.

REF	PROJECT	FUNDER / PROGRAMME	VALUE / PERIOD
P17	Synthetic methane plume dataset for satellite-based MR	EBRD Climate Change Innovation Programme	€30,000 · 2026–2027
P16	MethaneMapper: landfill methane plume detection	EBRD Climate Change Innovation Programme	€27,000 · 05/2025–11/2025
P15	Banana fields monitoring using remote sensing	EBRD Lebanon Innovation Programme	€30,000 · 03/2025–09/2025
P14	Winter wheat detection from Sentinel-2 imagery	ESA Network of Resources (NoR)	€5,000 · 03/2024–03/2025
P13	Qaraaoun Reservoir water monitoring	SEALACOM Researchers 2023	€4,500 · 01/2024–03/2024
P12	Buildings' rooftop segmentation using deep learning	EBRD Lebanon Innovation Programme	€5,000 · 12/2023–04/2024
P11	Updating the Lebanon solar rooftop potential map	ESA EO Data Access	3,378 km ² · 02/2023–07/2023
P10	AI and remote sensing in food security for the Arab World	AOAD	\$1,500 · 03/2022–04/2022
P09	Time-series analysis of Lebanese crops	ESA Network of Resources (NoR)	€5,000 · 11/2021–11/2022
P08	Grain bins classification for an 8MM AoI	SDXspace	\$7,000 · 04/2021–06/2021

Alongside these grant-funded programmes, RASID has delivered direct commercial engagements including a two-phase methane-monitoring project for a Brazilian firm, Arabic handwritten-OCR for the Iraqi public sector, iRAP road-safety automation (co-built with FRED Engineering), and free-flow speed modelling with the World Bank.

Team

A small, senior team spanning R&D, engineering, and commercial the same people who deliver every engagement.



01

Dr. Ali J. Ghandour

R&D Consultant — founding partner.



02

Reda Haidar

Commercial Director.



03

Hasan Nasrallah

AI Lead Engineer.



04

Hasan Wehbi

MLOps Engineer.



05

Mohamad El Moussawi

Full-Stack Developer.



06

Amira El Halabi

BD & Pre-Sales Engineer.

Let's talk.

RASID is ready to tailor a geospatial solution to your needs with flexible transaction/ subscription or consulting-based engagement models, backed by a decade of funded, delivered work.

EMAIL

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WEB

www.rasid.ai

PLATFORM

app.rasid.ai

BOOK A MEETING

calendly.com/rasid/30mins

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